

Fine-Tuning LLMs and RLHF

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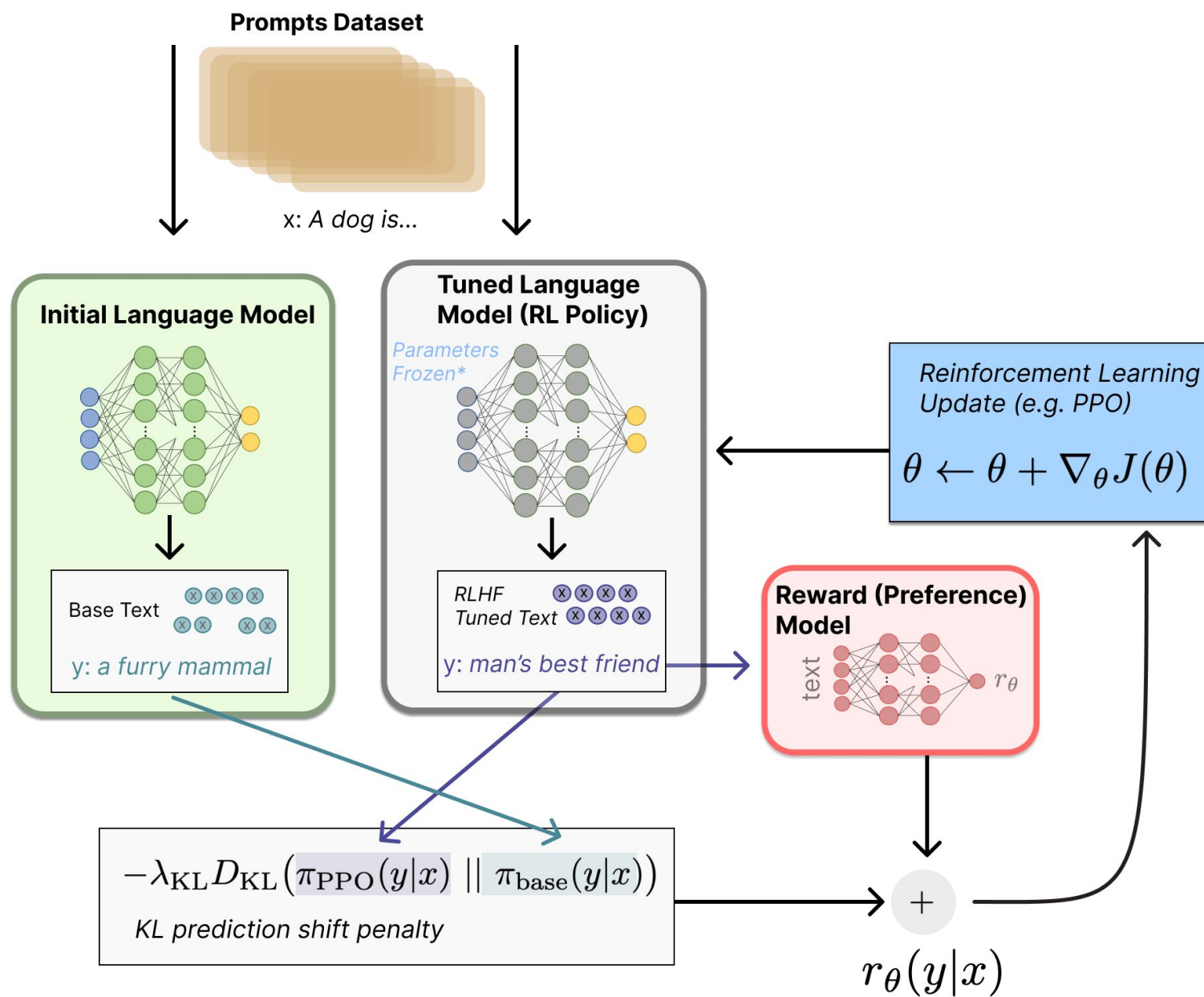


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- ▶ Pre-trained LLMs are powerful but not aligned to specific use-cases or human preferences.
- ▶ We need:
 - Adaptability (domain tuning)
 - Alignment (user intention)
 - Efficiency (cost-effective methods)
- ▶ Fine-tuning and RLHF bridge the gap between general intelligence and practical usability.

Motivating Examples:

- ▶ GPT-3 → InstructGPT
- ▶ LLaMA → Alpaca, Vicuna
- ▶ ChatGPT & Claude → RLHF-tuned

After this session, you will be able to:

- ▶ Explain and compare different fine-tuning methods for LLMs
- ▶ Understand the pipeline of Supervised Fine-Tuning and RLHF
- ▶ Describe PPO (Proximal Policy Optimization) and DPO (Direct Preference Optimization)
- ▶ Apply Low-Rank Adaptation (LoRA) and Quantized LoRA for efficient tuning
- ▶ Analyze trade-offs, limitations, and future directions of LLM adaptation

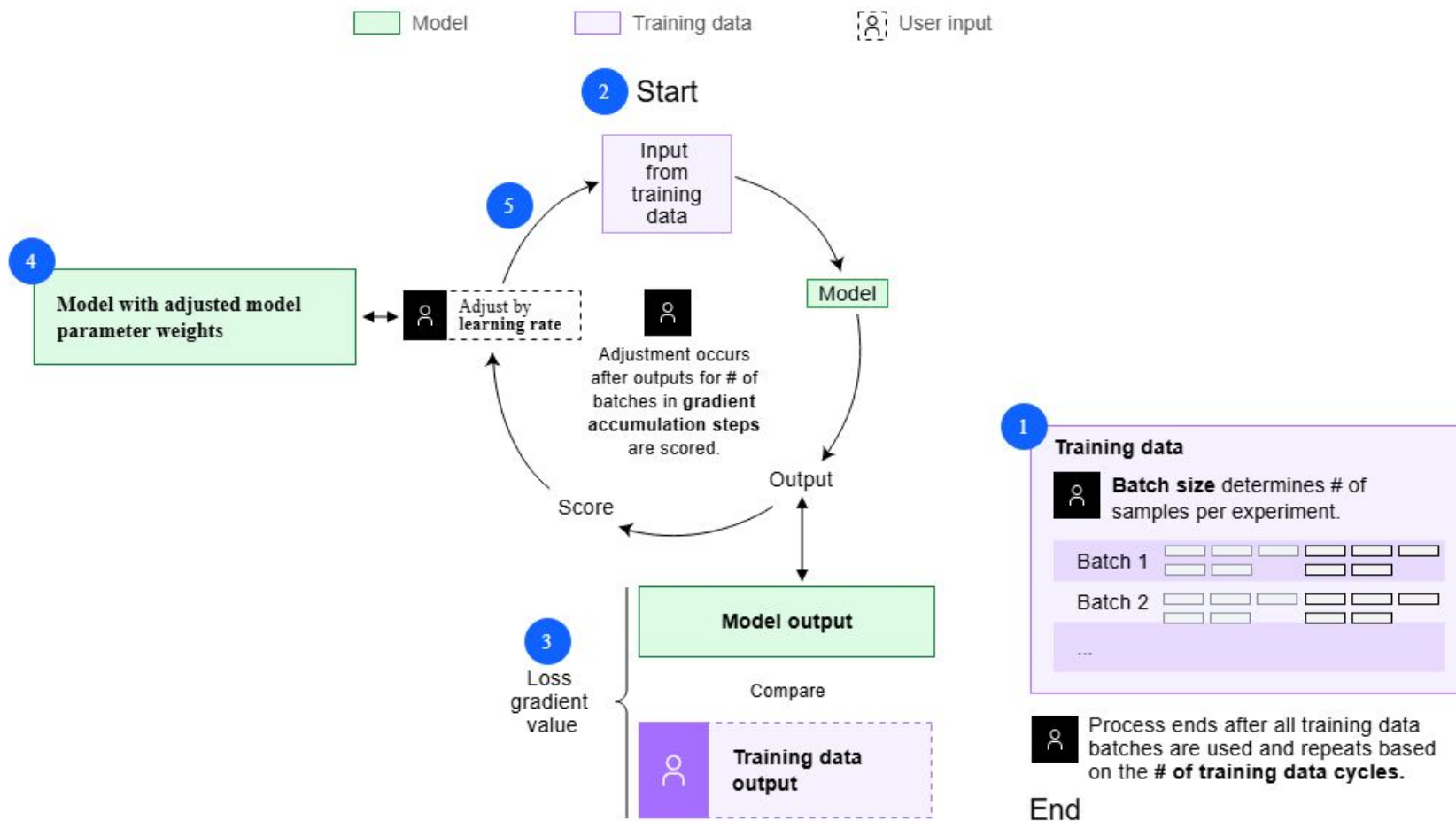
Fine-Tuning Methods for LLMs

Categories of Fine-Tuning

Method	Description	Parameters Updated	Efficiency
Full Fine-Tuning	Retrain all model weights	All	Expensive
Adapter Tuning	Add small bottlenecks (e.g., Houlsby)	Few	Efficient
Prefix Tuning	Tune soft prompts	Few tokens	Efficient
LoRA / QLoRA	Low-rank decomposition of weight deltas	Very few	Very Efficient
Instruction Tuning	Fine-tune on instruction-following datasets	All or partial	

- ▶ **Definition:** Fine-tune all parameters of the pre-trained model on downstream data.
- ▶ **Historical Use:** Common in early GPT-2 and BERT applications.
- ▶ **Pros:**
 - Maximum flexibility and performance
- ▶ **Cons:**
 - Expensive (requires large compute resources)
 - Prone to catastrophic forgetting
 - Not efficient for large models

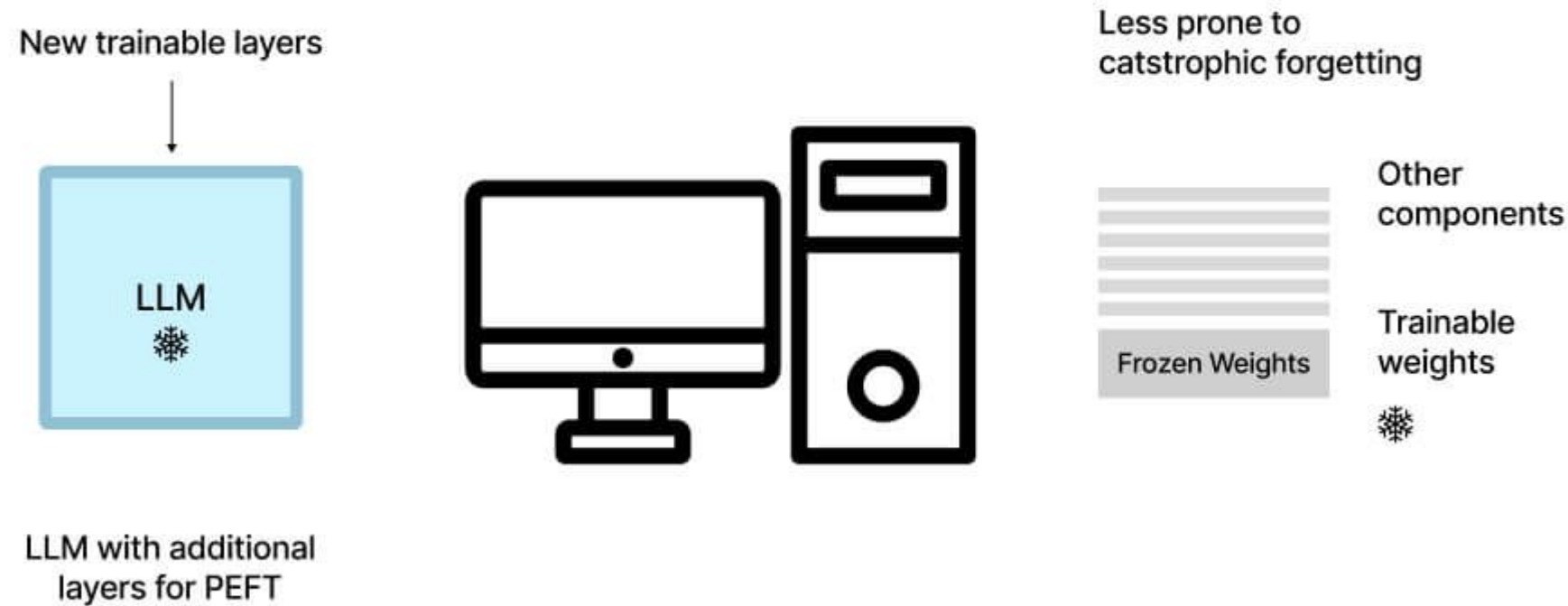
Full fine-tuning workflow





- ▶ **Key Idea:** Keep the base model frozen, tune only a small subset of parameters.
- ▶ **Types:**
 - Adapter Modules (Houlsby et al., 2019)
 - Prompt Tuning / Prefix Tuning
 - LoRA / QLoRA
- ▶ **Benefits:**
 - Enables multi-tasking and personalization
 - Suitable for low-resource adaptation
 - Reduces compute and memory requirements

Parameter efficient fine-tuning (PEFT)



Supervised Fine-Tuning (SFT)

What is Supervised Fine-Tuning (SFT)?

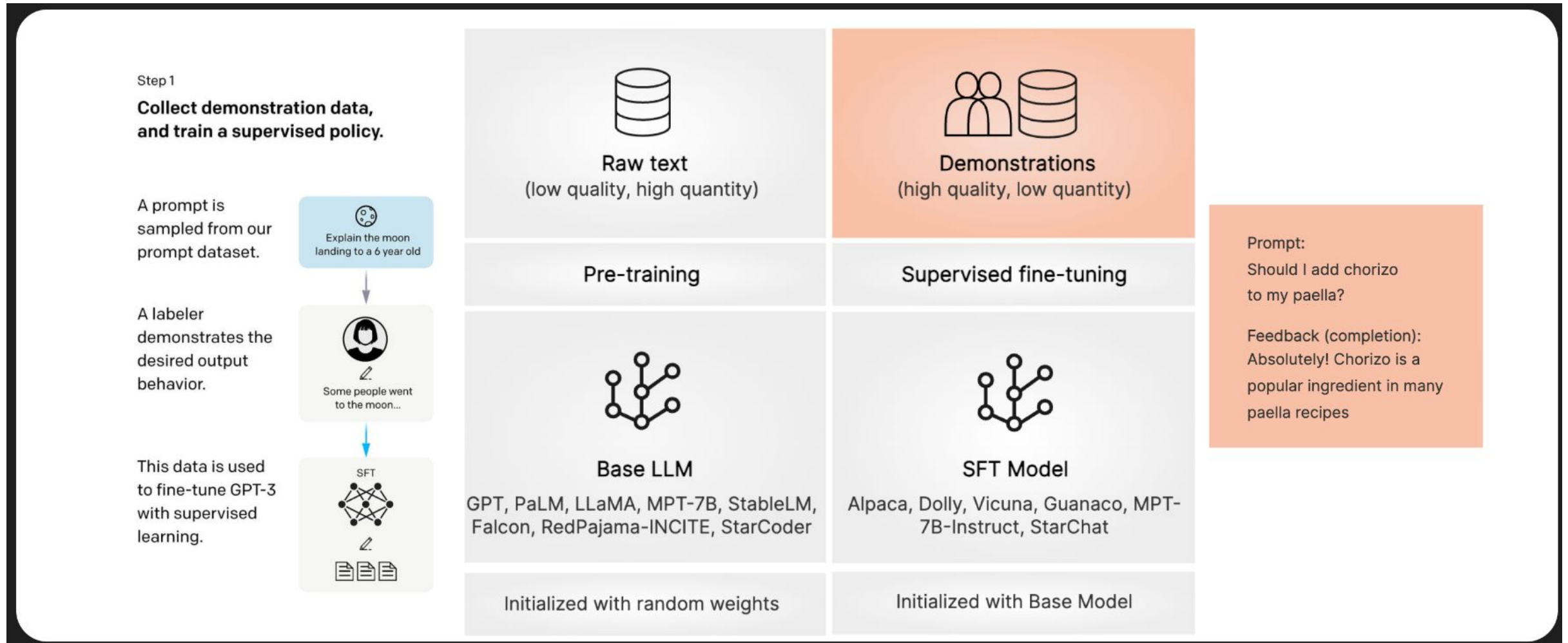
- ▶ **Definition:** SFT is training on labeled instruction-response pairs.
- ▶ **Example:**
 - *Prompt:* “Explain black holes to a 5-year-old”
 - *Response:* “Black holes are like big vacuum cleaners in space. . .”
- ▶ **Objective:** Optimize the log-likelihood of the correct response given the prompt.

Loss Function:

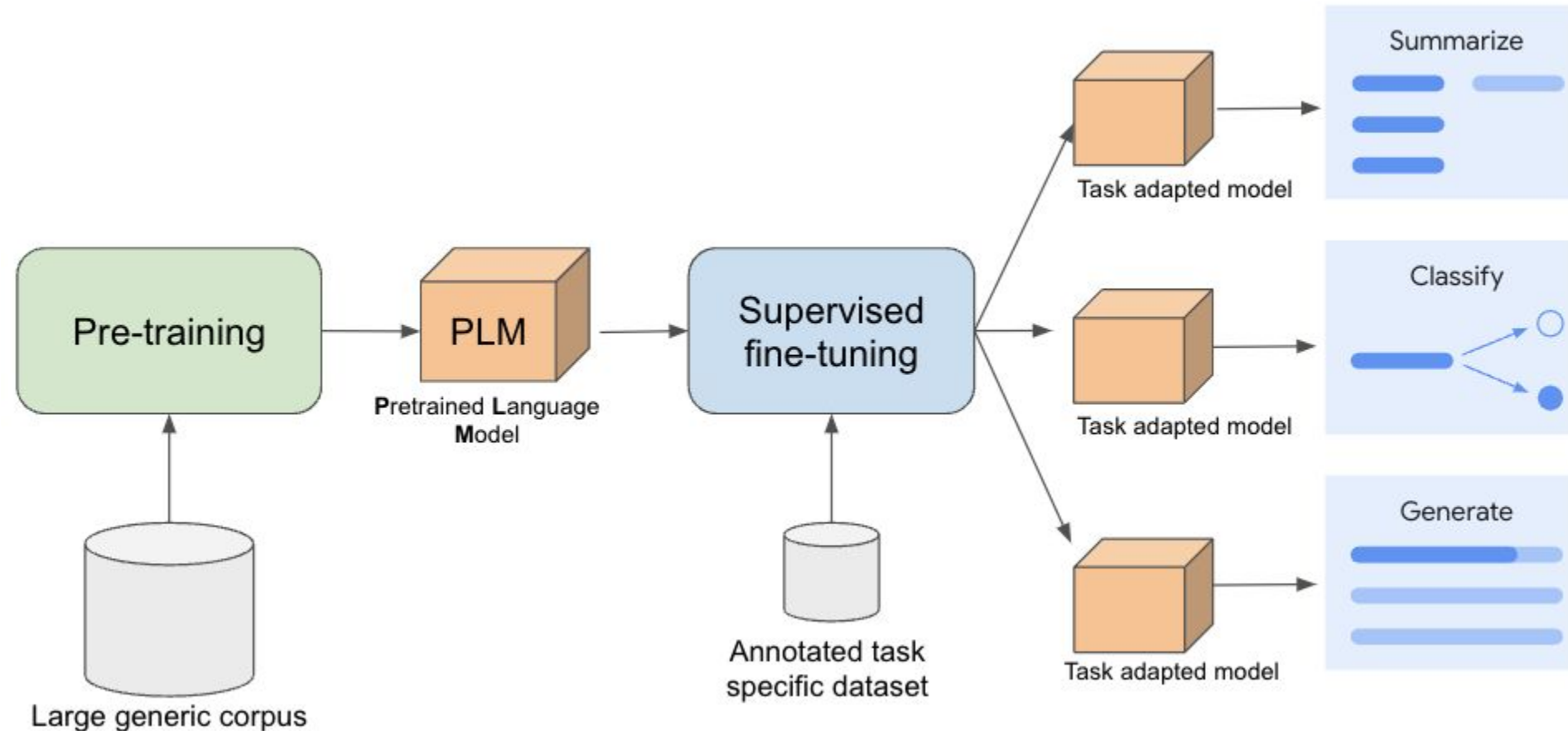
$$L_{\text{SFT}} = - \sum_{t=1}^T \log p_{\theta}(y_t \mid y_{<t}, x)$$

where x is the prompt, y is the response, and T is the response length.

What is Supervised Fine-Tuning (SFT)?



Supervised Fine Tuning for Gemini LLM



Supervised Fine Tuning for Gemini LLM



► Instructional datasets:

- OpenAI: InstructGPT (Anthropic Helpfulness data)
- Stanford Alpaca (52k GPT-3 generated instructions)
- Dolly, ShareGPT, OASST, UltraChat

► **Note:** Quality of supervision greatly affects model behavior.



- ▶ Cannot capture nuanced human preferences or values.
- ▶ May reinforce existing biases or hallucinations present in the data.
- ▶ Risk of overfitting, especially on synthetic or noisy datasets.
- ▶ Often leads to safe but bland and generic responses.

RLHF — Reinforcement Learning with Human Feedback

Why RLHF?

- ▶ SFT doesn't teach the model what humans prefer.
- ▶ RLHF introduces reward learning and policy optimization.
- ▶ Inspired by InstructGPT (Ouyang et al., 2022).

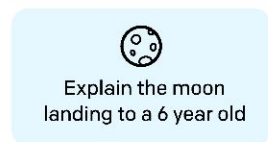
1. **Supervised Fine-Tuning (SFT):** Train the base model on instruction-response pairs.
2. **Reward Model (RM) Training:** Learn a reward function from human preference rankings.
3. **RL Optimization (e.g., PPO):** Optimize the language model to maximize the learned reward.

RLHF Pipeline Overview

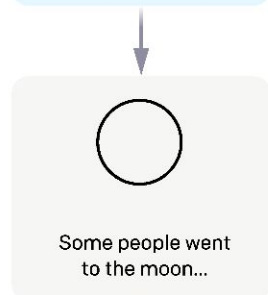
Step 1

**Collect demonstration data,
and train a supervised policy.**

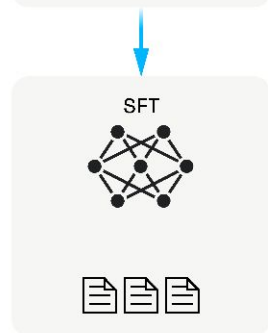
A prompt is
sampled from our
prompt dataset.



A labeler
demonstrates the
desired output
behavior.



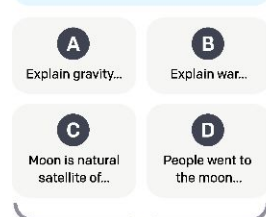
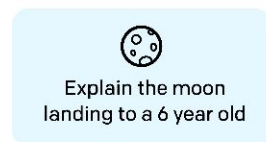
This data is used
to fine-tune GPT-3
with supervised
learning.



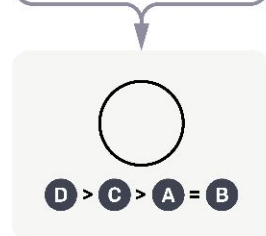
Step 2

**Collect comparison data,
and train a reward model.**

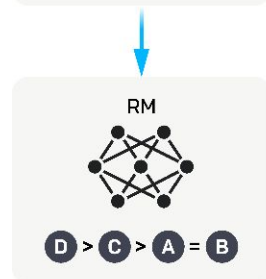
A prompt and
several model
outputs are
sampled.



A labeler ranks
the outputs from
best to worst.



This data is used
to train our
reward model.



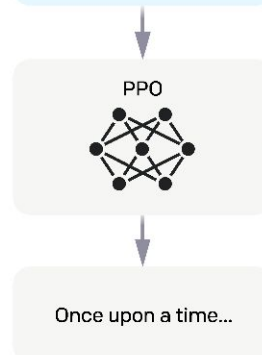
Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**

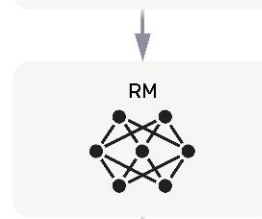
A new prompt
is sampled from
the dataset.



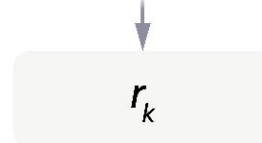
The policy
generates an output.



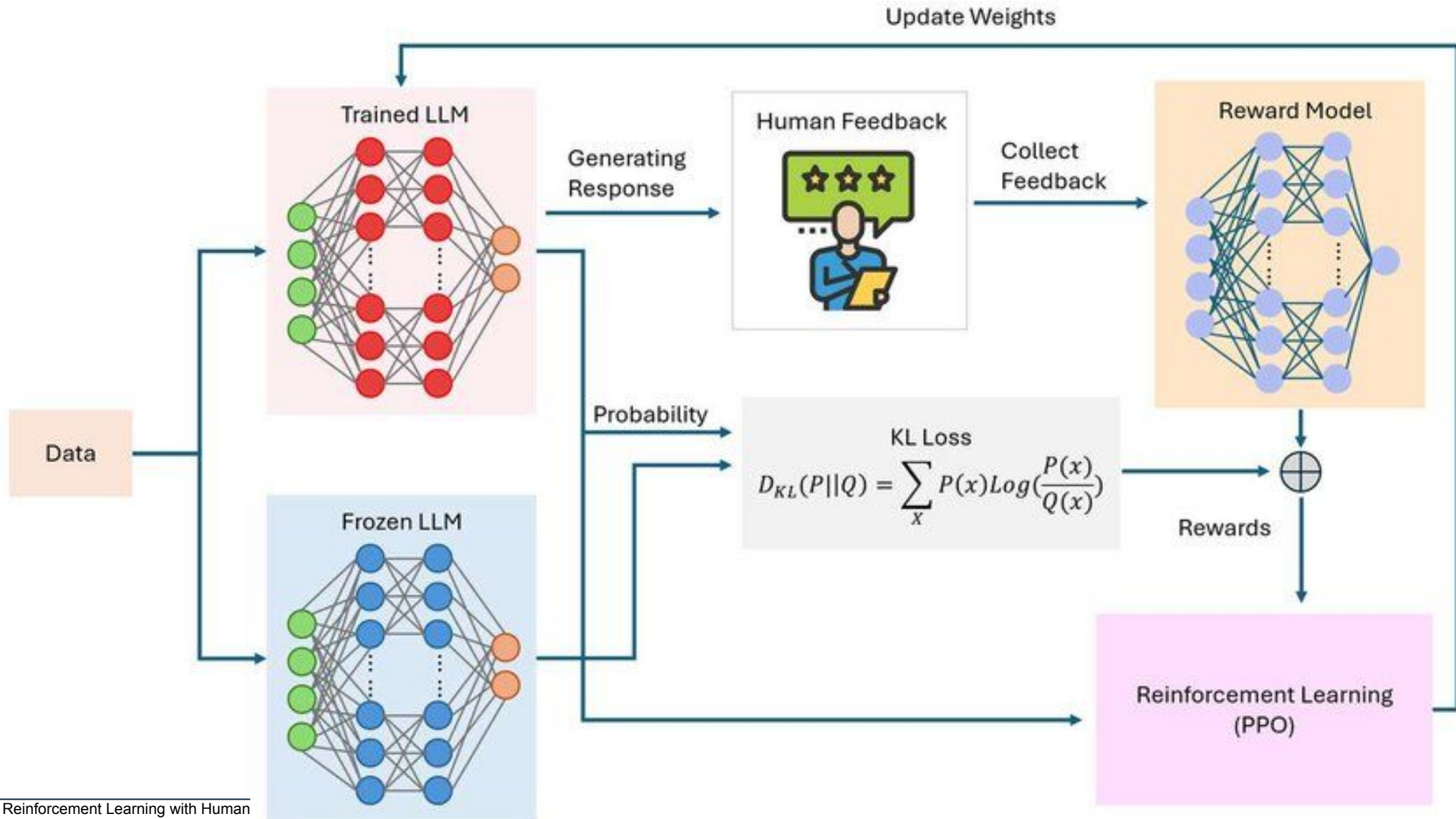
The reward model
calculates a
reward for
the output.



The reward is
used to update
the policy
using PPO.

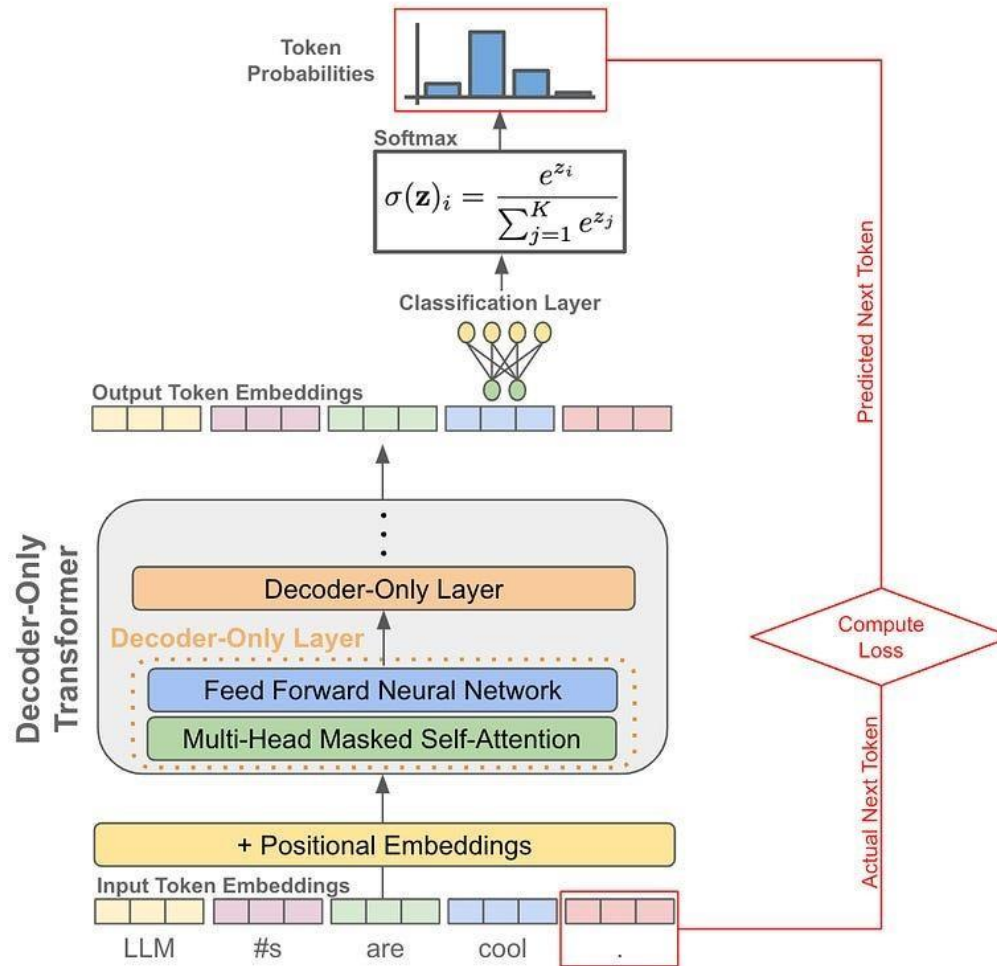


RLHF Pipeline

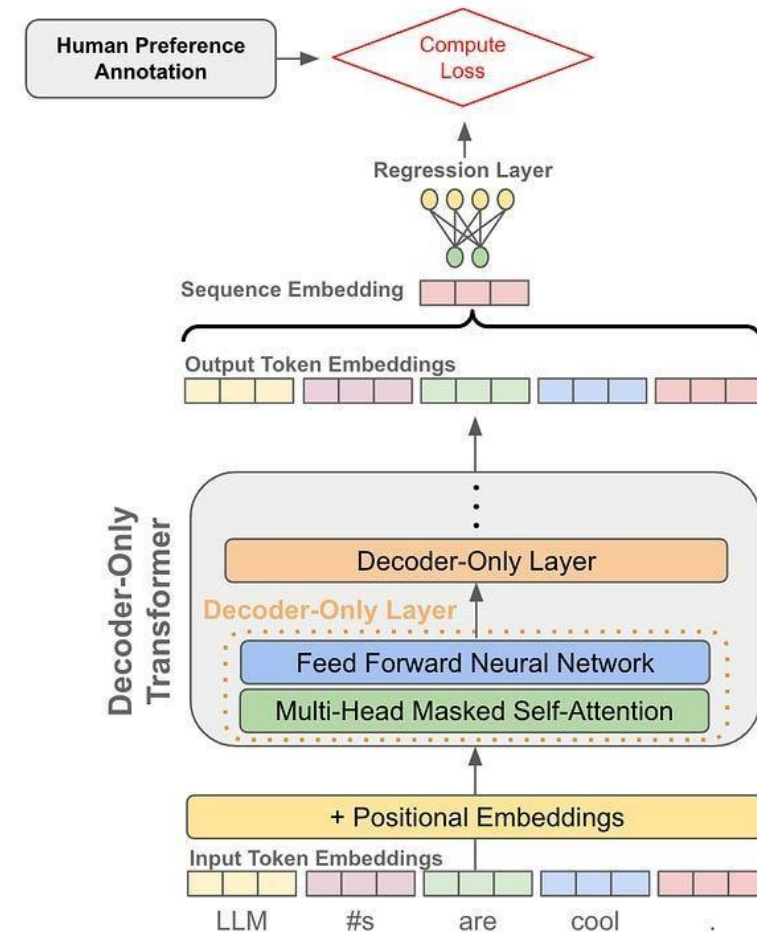


Model Fine-tuning for RLHF

Next-Token Prediction with an LLM



Reward Model Structure



- ▶ **Given:** Two responses y_A and y_B to the same prompt.
- ▶ **Human Feedback:** Annotator selects which response is preferred.
- ▶ **Learning:** The reward model r_ϕ is trained to assign higher scores to preferred responses.
- ▶ **Pairwise Loss:**

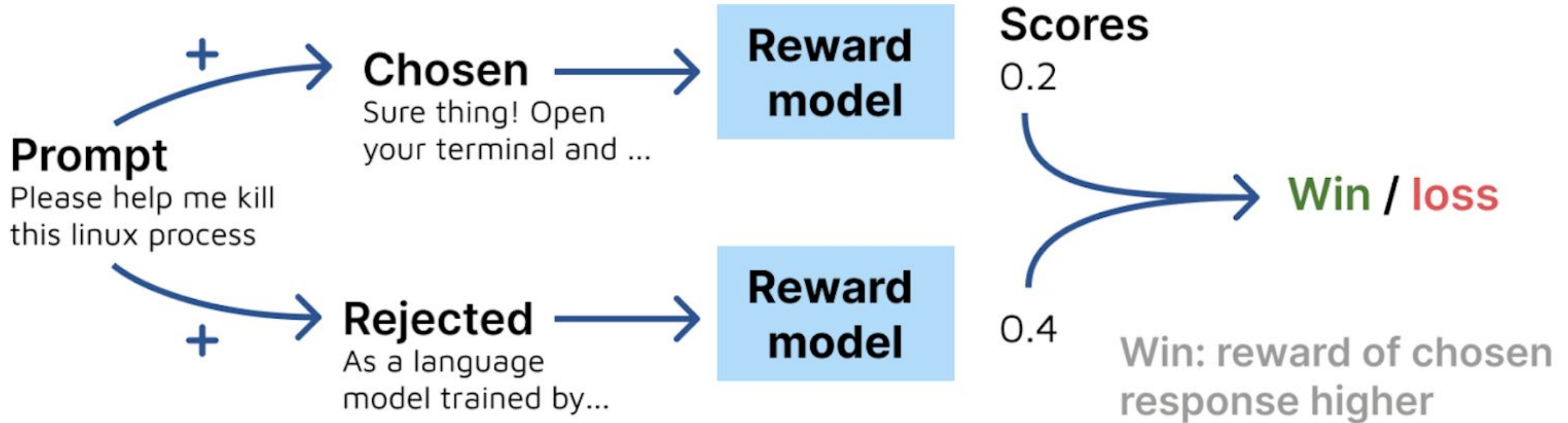
$$L_{\text{RM}} = -\log \sigma(r_\phi(y_A) - r_\phi(y_B))$$

where σ is the sigmoid function.

- ▶ The RM acts as a proxy for human judgment in downstream RL optimization.

Reward Model (RM)

Manually curated preferences



Prompts to test capabilities



- ▶ **Objective:** Maximize the reward given by the reward model (RM).
- ▶ **Algorithm:** Proximal Policy Optimization (PPO) is used to update the language model.
- ▶ **PPO Loss:**

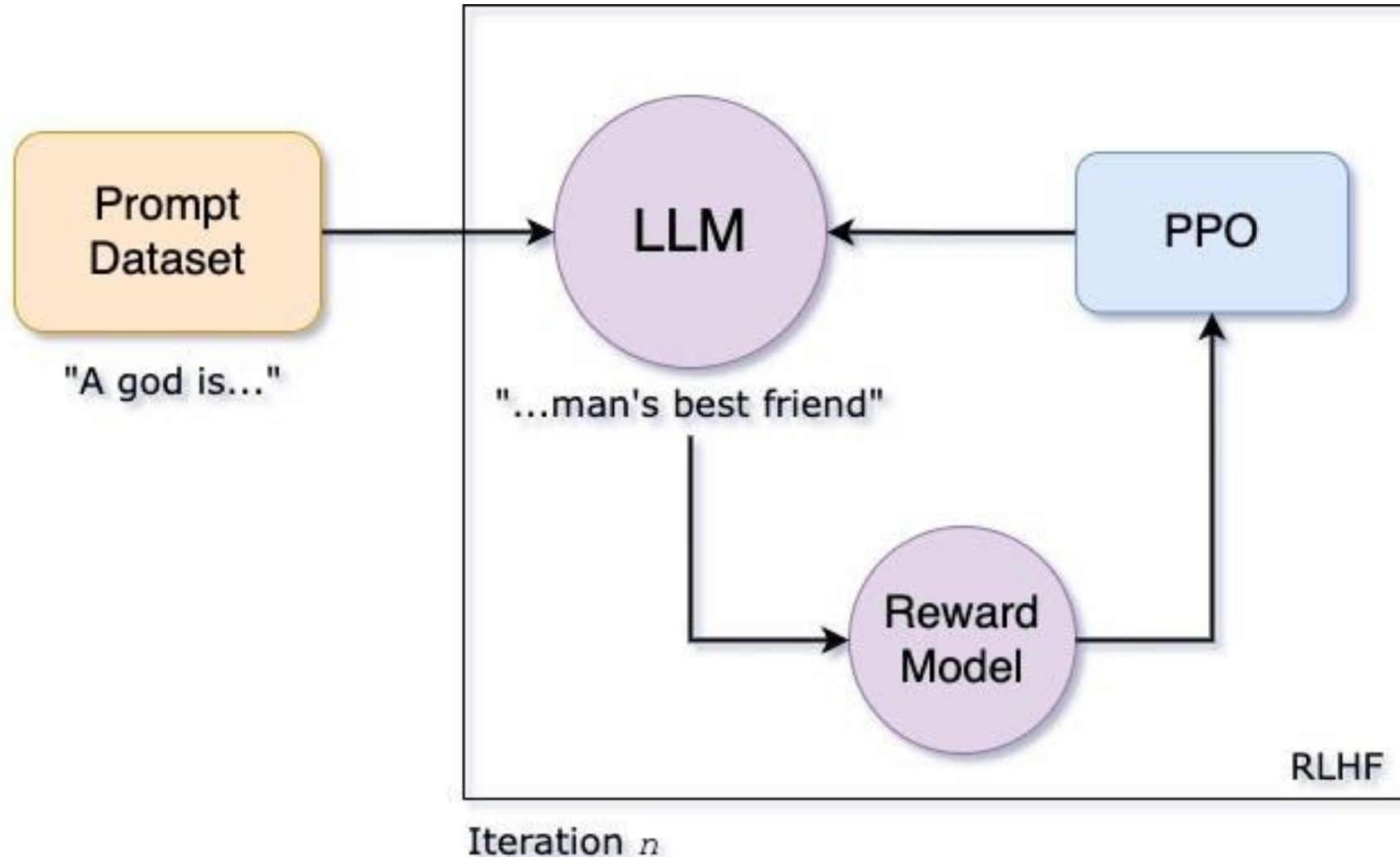
$$L_{\text{PPO}} = \mathbb{E} \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip} (r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

where $r_t(\theta) = \frac{\pi_{\theta}(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}$ is the probability ratio, and $\hat{A}_t$ is the advantage estimate.

- ▶ **KL Penalty:** A KL-divergence penalty is added to keep the policy close to the base (SFT) model:

$$L_{\text{KL}} = \beta \text{KL} (\pi_{\theta} \parallel \pi_{\text{SFT}})$$

RLHF Training: Proximal Policy Optimization (PPO)



- ✓ Stable optimization
- ✓ Encourages exploration
- ✗ Expensive (needs many rollouts)
- ✗ Sensitive to reward model errors

Direct Preference Optimization (DPO)

- ▶ **New method:** Bypasses RL and reward model training entirely.
- ▶ **Key Idea:** Optimize the policy directly from human preference data.
- ▶ **Reference:** Rafailov et al., 2023.

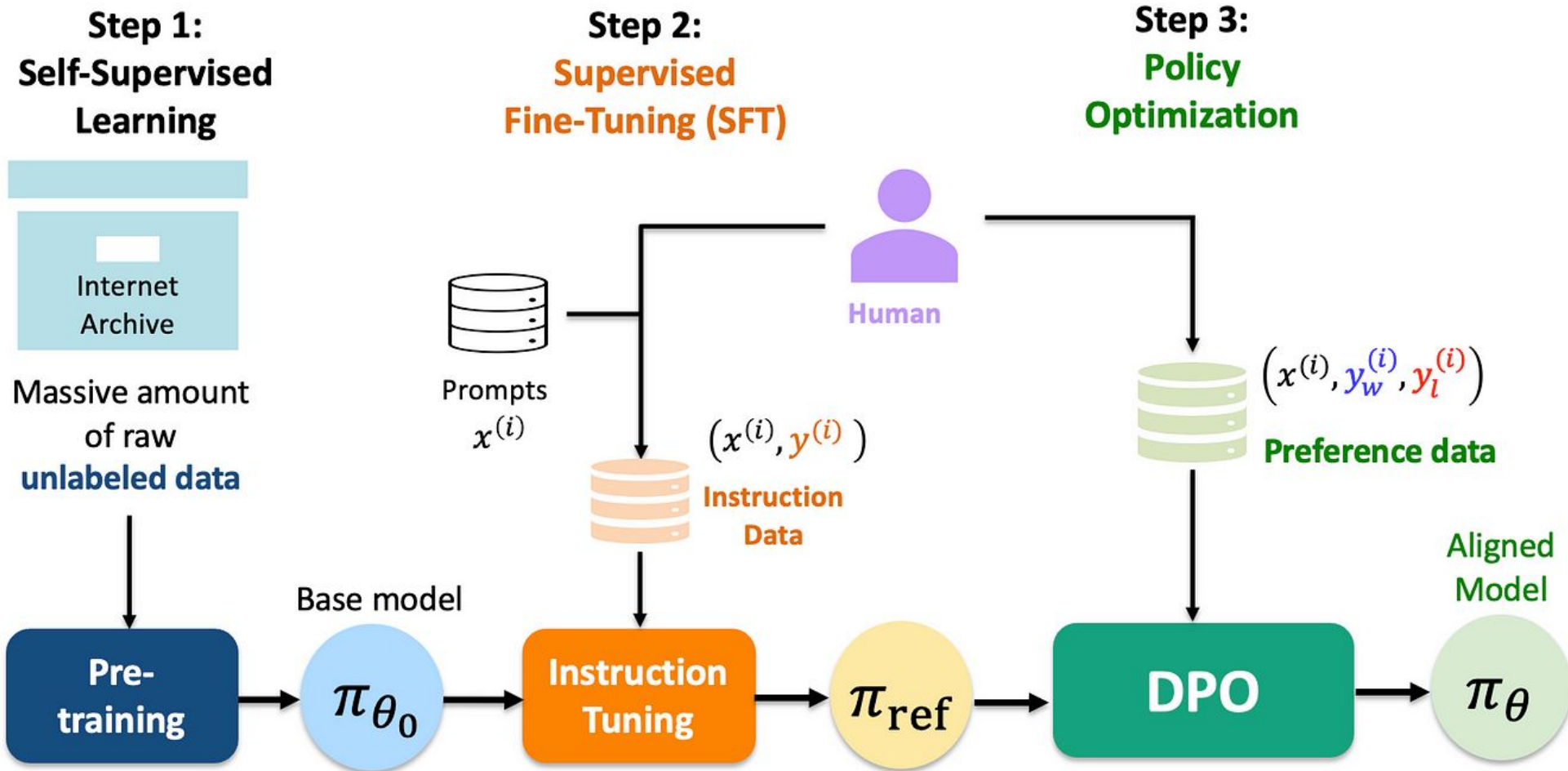
DPO Loss:

$$L_{\text{DPO}} = \log \frac{\exp \left(\beta \log \frac{\pi(y^+)}{\pi_0(y^+)} \right)}{\exp \left(\beta \log \frac{\pi(y^+)}{\pi_0(y^+)} \right) + \exp \left(\beta \log \frac{\pi(y^-)}{\pi_0(y^-)} \right)}$$

where y^+ is the preferred response, y^- is the dispreferred response, π is the current policy, π_0 is the reference (SFT) policy, and β is a temperature parameter.

- ▶ **Benefits:**
 - Simpler and more stable than PPO
 - No need for reward model training or RL rollouts

Direct Preference Optimization (DPO)



LoRA and Quantized LoRA

What is LoRA?

- ▶ **Key Idea:** Update low-rank matrices instead of full weights.
- ▶ For a weight matrix $W \in \mathbb{R}^{d \times k}$:

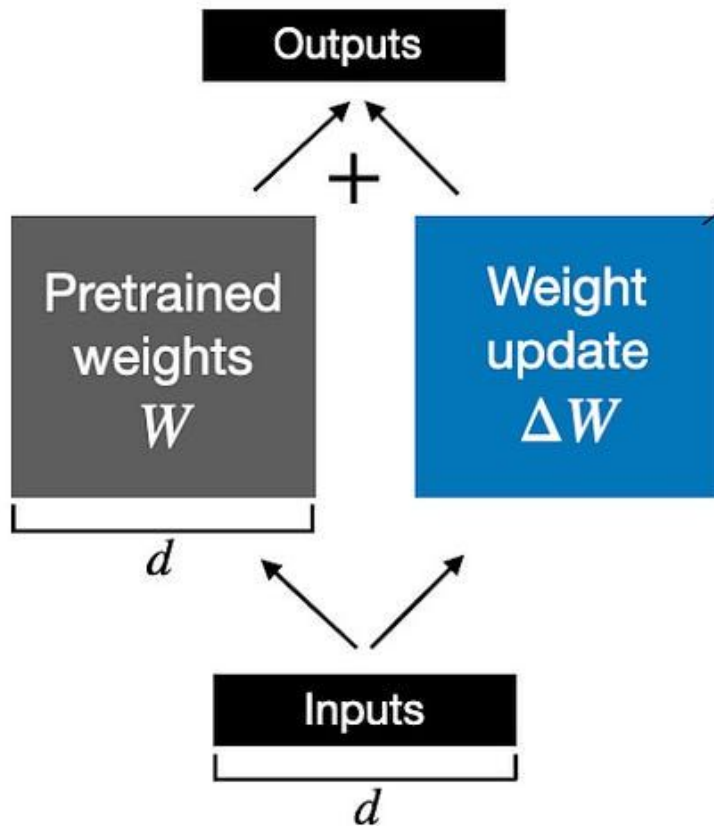
$$W' = W + \Delta W, \quad \Delta W = AB^T$$

where $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{k \times r}$.

- ▶ Only train $A, B \rightarrow$ drastically reduces parameters.

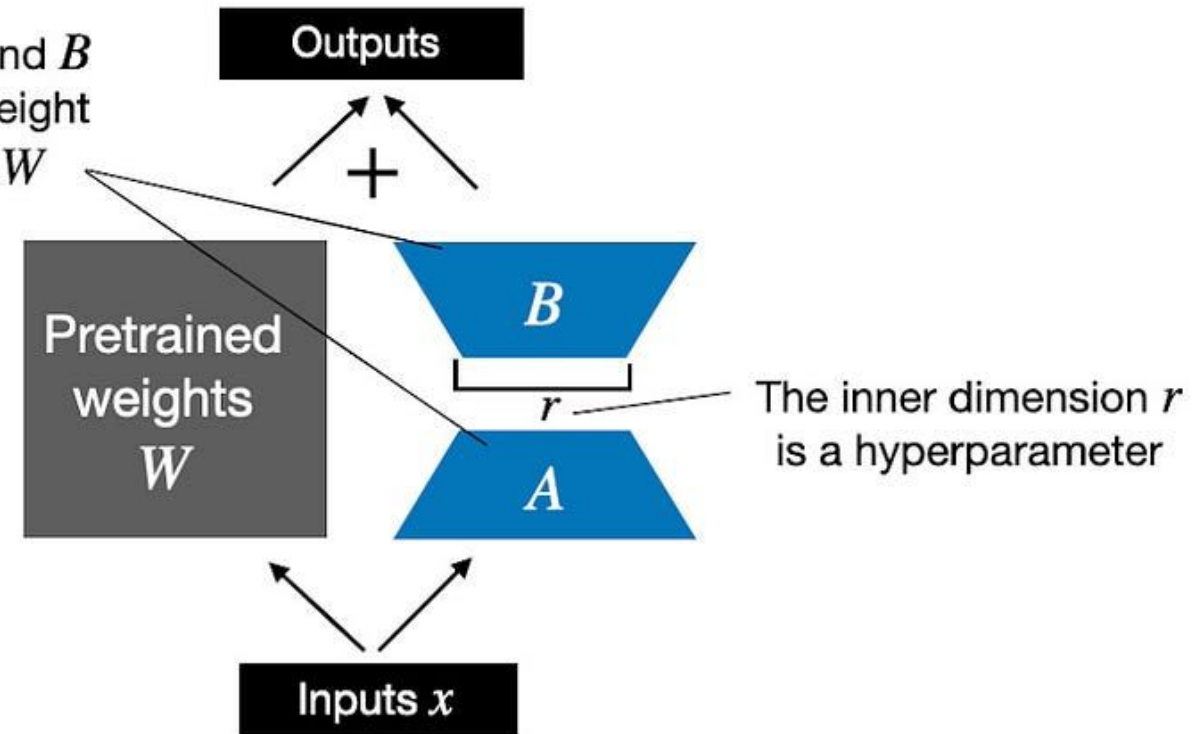
What is LoRA?

Weight update in regular finetuning



LoRA matrices A and B
approximate the weight
update matrix ΔW

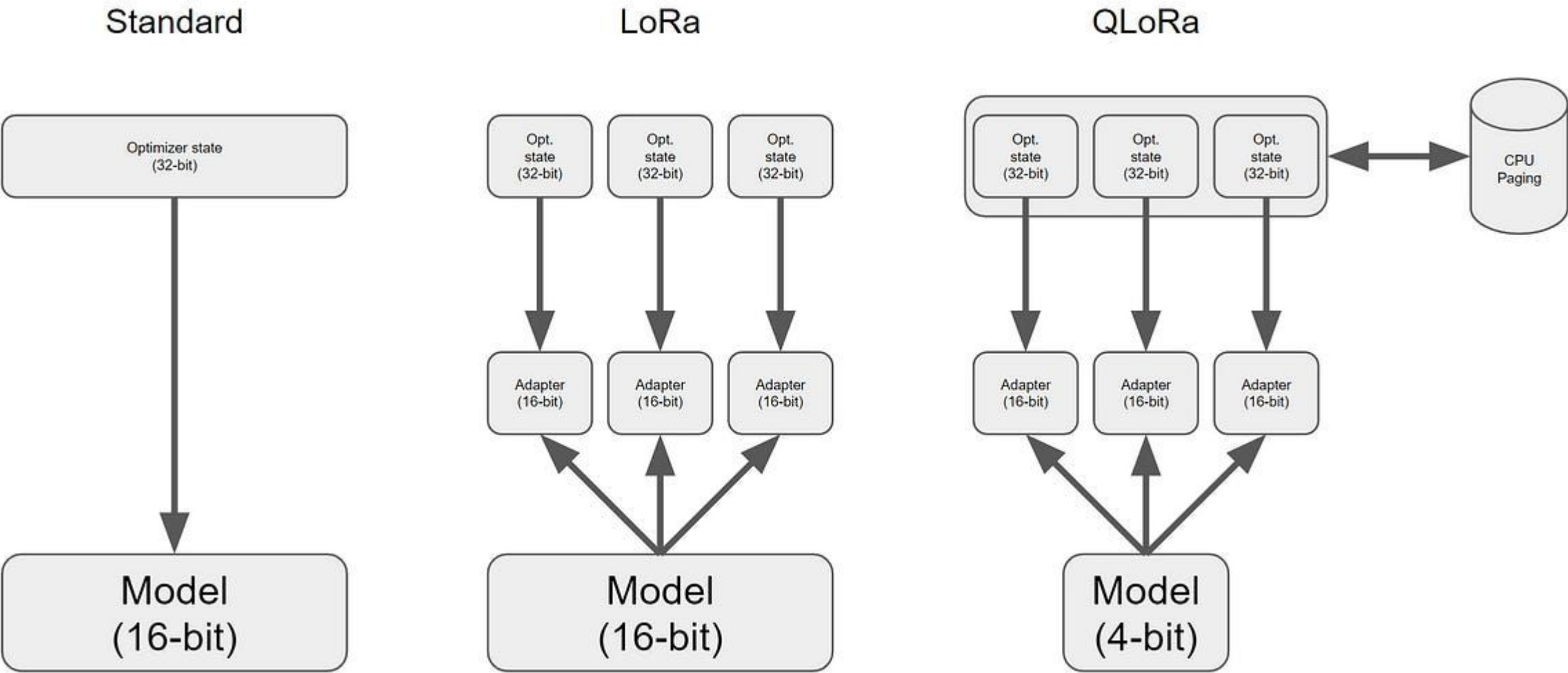
Weight update in LoRA



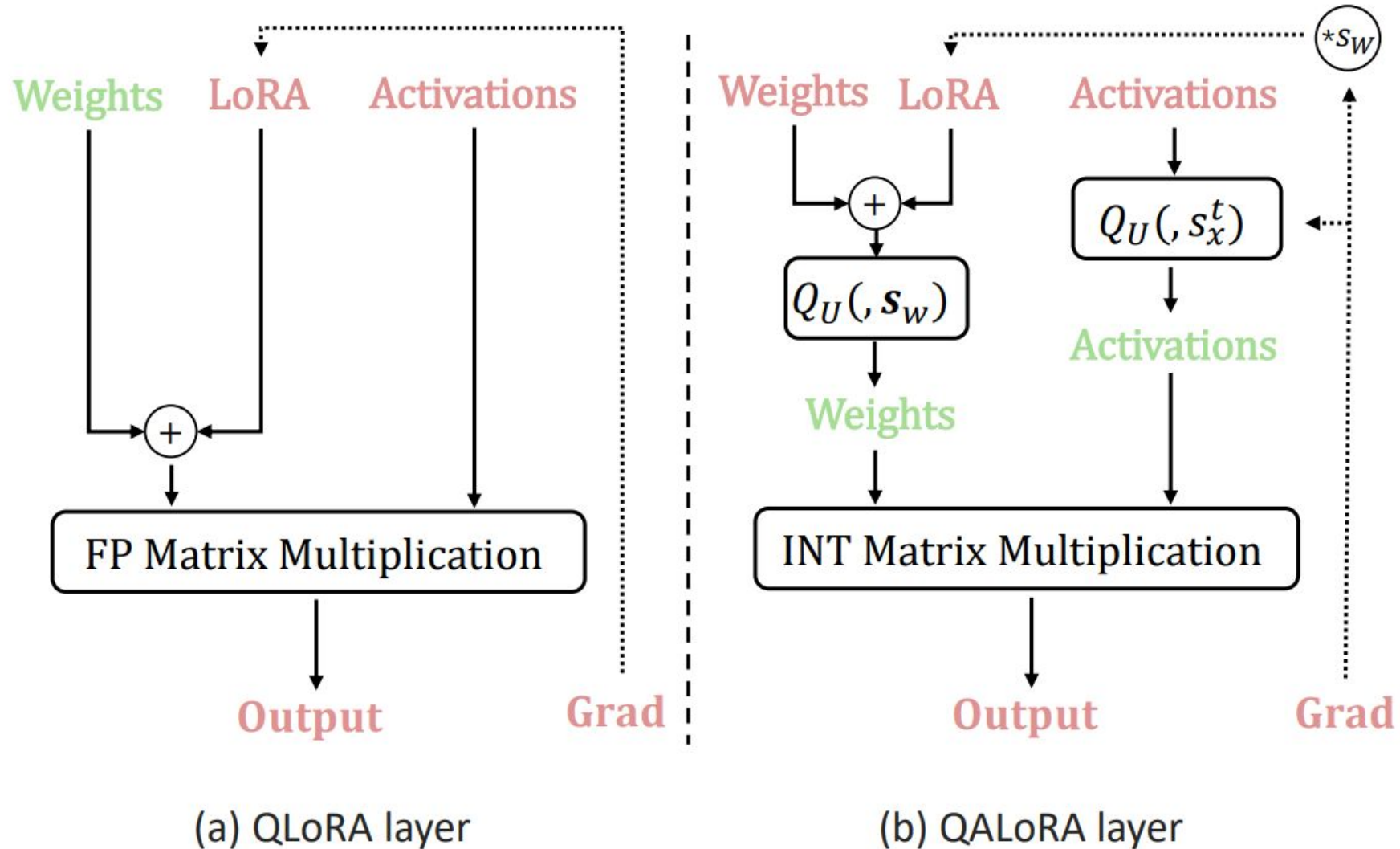
- ▶ **Very lightweight:** Only 0.1%–1% of parameters are trainable.
- ▶ **Hardware friendly:** Enables fine-tuning on consumer GPUs and even laptops.
- ▶ **Modular:** Supports plug-and-play adapters for different tasks or users.
- ▶ **Personalization:** Allows efficient user- or domain-specific adaptation.

- ▶ **Key Idea:** Fine-tune large language models in 4-bit precision using LoRA adapters.
- ▶ **Scalability:** Enables fine-tuning of 65B parameter models on a single 48GB GPU.
- ▶ **Efficiency:** Highly memory-efficient with no significant performance loss (Dettmers et al., 2023).
- ▶ **Techniques:**
 - Double quantization
 - Paged optimizers
 - NF4 (NormalFloat 4-bit) quantization format

QLoRA — Quantized LoRA



QLoRA — Quantized LoRA



► **Instruction tuning for resource-constrained settings:**

- Enables fine-tuning large models on modest hardware (e.g., consumer GPUs, laptops).
- Used for domain adaptation, personalization, and rapid prototyping.

► **Popular fine-tuned models:**

- **Alpaca, Guanaco, Vicuna, Mistral:** All leverage LoRA/QLoRA for efficient instruction tuning.

► **Model merging and compositionality:**

- Merge multiple LoRA adapters for multi-domain or multi-task capabilities.
- Compose adapters for new tasks without retraining the base model.

Limitations of Fine-Tuning and RLHF

► RLHF:

- Reward models can be misaligned with true human preferences.
- Training loops are computationally expensive and complex.

► DPO:

- Still depends on high-quality human preference data.

► LoRA:

- Only updates a small subset of parameters; may miss global interactions.

► Quantization:

- Requires careful selection and tuning of quantization formats (e.g., NF4).

- ▶ **RLAIF (RL with AI Feedback):** Automate human labelers using strong AI models for preference data.
- ▶ **Constitutional AI:** Use rule-based alignment and self-critique to guide model behavior (e.g., Anthropic's approach).
- ▶ **Open RLHF Datasets:** Promote better community sharing and reproducibility with open preference datasets.
- ▶ **Multi-modal Alignment:** Extend alignment techniques to vision, audio, and code generation tasks.
- ▶ **Reward Hacking Defense:** Develop robust reward functions to prevent models from exploiting reward loopholes.

- ▶ Fine-tuning enables specialization and alignment of LLMs.
- ▶ Supervised Fine-Tuning (SFT) is the foundational step.
- ▶ RLHF incorporates human preference alignment using PPO.
- ▶ DPO streamlines the process with a direct preference loss.
- ▶ LoRA and QLoRA make adaptation efficient and accessible.
- ▶ The field is rapidly evolving with ongoing innovation.

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Credits

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